

Universidad Peruana de Ciencias Aplicadas



INFORME DE TRABAJO FINAL

Big Data

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Introduction

This project focuses on building a data intelligence pipeline for the healthcare sector. Using a synthetic dataset of 55,500 patient records, we aim to implement a complete lifecycle from automated ingestion to advanced entity clustering and graph-based medical analysis.

Problem Statement

The core challenge is the lack of structured, high-integrity data in hospital management. This project addresses the need for a "Catalog Layer" that cleans raw data, standardizes medical categories, and handles high-cardinality features, providing a foundation for predictive clinical insights.

Product Question

Is it possible to cluster patient profiles and treatment pathways to identify patterns between medical conditions, medication types, and hospital admission outcomes?

Source Inventory

- **Primary Source:** healthcare_dataset.csv (Synthetic Healthcare Data).
- **Entity Count:** Patients (unique individuals), Doctors, and Healthcare Providers.
- **Key Features:** Medical Condition, Admission Type, Medication, and Test Results.
- **Temporal Scope:** Continuous tracking from Admission Date to Discharge Date.

Dataset url	https://www.kaggle.com/datasets/prasad22/healthcare-dataset
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Schema Draft

Entity Tables

The data model is composed of multiple entity tables derived from the original dataset. Each table represents a key component of the healthcare domain and is designed to support relational and analytical operations.

The main entities include:

- **patients:** stores demographic information about each patient, such as age, gender, and blood type.
- **hospitals:** contains identifiers and names of healthcare institutions.
- **doctors:** includes information about attending medical professionals.
- **medical_conditions:** represents diagnosed health conditions.
- **medications:** represents prescribed treatments.
- **medical_records:** serves as the central table that captures each clinical event, including patient interactions, diagnosis, treatment, and hospital-related data.

Keys (Primary and Foreign Keys)

The schema uses primary keys (PK) to uniquely identify each record within a table, and foreign keys (FK) to establish relationships between tables.

Primary Keys (PK):

- patients.patient_id
- hospitals.hospital_id
- doctors.doctor_id
- medical_conditions.condition_id
- medications.medication_id
- medical_records.record_id

Foreign Keys (FK):

- medical_records.patient_id → patients.patient_id
- medical_records.hospital_id → hospitals.hospital_id
- medical_records.doctor_id → doctors.doctor_id
- medical_records.condition_id → medical_conditions.condition_id
- medical_records.medication_id → medications.medication_id

Relationships and Joins

The schema defines one-to-many relationships between entity tables and the central **medical_records** table.

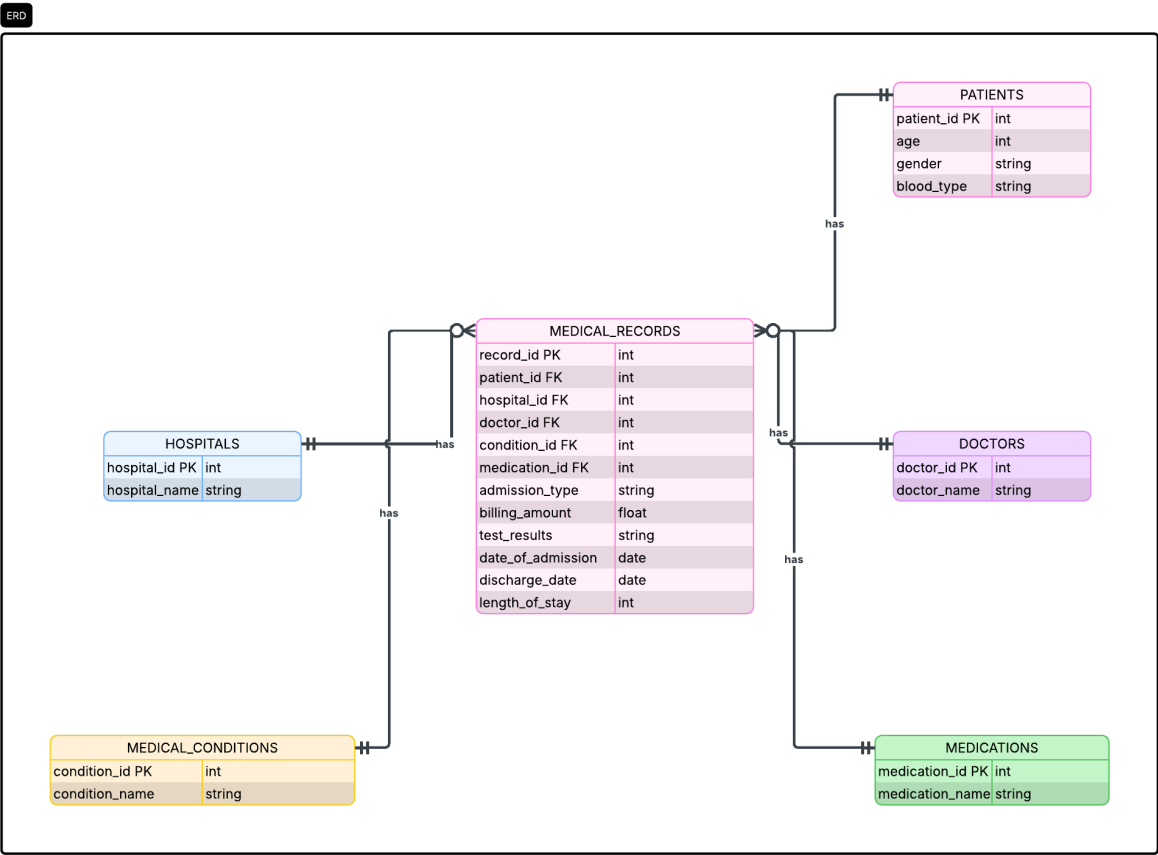
- One patient can have multiple medical records.
- One hospital can have multiple medical records.
- One doctor can be associated with multiple medical records.
- One medical condition can appear in multiple records.
- One medication can appear in multiple records.

These relationships are implemented through foreign keys and enable efficient joins for analytical purposes.

Example joins include:

- patients ↔ medical_records (via patient_id)
- hospitals ↔ medical_records (via hospital_id)
- medical_conditions ↔ medical_records (via condition_id)
- medications ↔ medical_records (via medication_id)

ERD



The schema transforms the original dataset into a normalized relational model centered on medical_records. Key entities such as patients, hospitals, doctors, conditions, and medications are separated into individual tables and connected through foreign keys, enabling scalable analysis, clustering, and graph-based modeling.

Data Dictionary

Raw Dataset Overview

Column Name	Description	Data Type
Name	Patient's name	string
Age	Age of the patient	integer
Gender	Gender of the patient (Male, Female)	categorical
Blood Type	Blood type of the patient	categorical

Medical Condition	Diagnosed medical condition	categorical
Date of Admission	Date of patient admission	datetime
Doctor	Attending doctor's name	string
Hospital	Hospital name	string
Insurance Provider	Insurance company	categorical
Billing Amount	Total billing amount	float
Room Number	Room assigned to the patient	integer
Admission Type	Type of admission (Emergency, Elective, Urgent)	categorical
Discharge Date	Date of patient discharge	datetime
Medication	Prescribed medication	categorical
Test Results	Results of medical tests	categorical

The raw dataset was transformed into multiple normalized tables to support relational analysis, clustering, and graph-based modeling.

Tables Description

Table: Patients

The patients table contains demographic information for each individual in the dataset. It includes attributes such as age, gender, and blood type, and each patient is uniquely identified by a `patient_id`. This table serves as a core entity for analyzing patient-level patterns.

Column	Type	Description
<code>patient_id</code>	int	unique identifier for each patient

age	int	age of the patient
gender	categorical	patient gender
blood_type	categorical	blood type of the patient

Table: Hospitals

The hospitals table stores information about healthcare institutions where patients receive treatment. Each hospital is uniquely identified by a `hospital_id` and is linked to medical records through relational keys.

Column	Type	Description
hospital_id	int	unique identifier for each hospital
hospital_name	string	name of the hospital

Table: Doctors

The doctors table contains information about medical professionals responsible for patient care. Each doctor is uniquely identified by a `doctor_id` and may be associated with multiple medical records.

Column	Type	Description
doctor_id	int	unique identifier for each doctor
doctor_name	string	name of the doctor

Table: medical_conditions

The `medical_conditions` table represents the set of diagnosed health conditions. Each condition is assigned a unique `condition_id`, enabling consistent referencing and reducing redundancy across the dataset.

Column	Type	Description
condition_id	int	unique identifier for the condition

condition_name	categorical	name of the medical condition
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Table: Medications

The medications table includes the different treatments or drugs prescribed to patients. Each medication is uniquely identified by a medication_id and is linked to medical records for analysis of treatment patterns.

Column	Type	Description
medication_id	int	unique identifier for the medication
medication_name	categorical	name of the medication

Table: medical_records

The medical_records table is the central fact table of the dataset. It captures each clinical event, including patient interactions with hospitals and doctors, diagnoses, treatments, admission details, and billing information. This table connects all other entities through foreign keys and enables comprehensive analytical tasks such as clustering, pattern discovery, and graph-based modeling.

Column	Type	Description
record_id	int	unique identifier for each record
patient_id	int	reference to patients table
hospital_id	int	reference to hospitals table
doctor_id	int	reference to doctors table
condition_id	int	reference to medical_conditions table
medication_id	int	reference to medications table
admission_type	categorical	type of admission (Emergency, Elective, Urgent)
billing_amount	float	total cost of treatment
test_results	categorical	results of medical tests

date_of_admission	datetime	admission date
discharge_date	datetime	discharge date
length_of_stay	int	number of days hospitalized

Ethics and Access Note

- **Data Origin:** The data is a synthetic dataset designed for educational and data science research.
- **Usage Justification:** We are utilizing this dataset under academic license for the "Domain Discovery, Recommendation, and Graph Intelligence" course.
- **Personal Data Risks and Mitigation:** While the data contains "Name" fields, they are synthetically generated and do not correspond to real individuals (PII). To prevent any accidental correlation, these identifiers will be treated as non-functional tokens or excluded from the final feature vectors.

Preprocessing and feature engineering

Preprocessing

Data Loading and Initial Inspection

The dataset, `healthcare_dataset.csv`, was loaded into a Pandas DataFrame. An initial inspection using `df.info()` revealed 15 columns with various data types, primarily object (strings), int, and float.

```

1 df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55500 entries, 0 to 55499
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Name                  55500 non-null  object
 1   Age                   55500 non-null  int64
 2   Gender                55500 non-null  object
 3   Blood Type            55500 non-null  object
 4   Medical Condition     55500 non-null  object
 5   Date of Admission     55500 non-null  object
 6   Doctor                55500 non-null  object
 7   Hospital              55500 non-null  object
 8   Insurance Provider    55500 non-null  object
 9   Billing Amount         55500 non-null  float64
10  Room Number           55500 non-null  int64
11  Admission Type         55500 non-null  object
12  Discharge Date        55500 non-null  object
13  Medication             55500 non-null  object
14  Test Results          55500 non-null  object
dtypes: float64(1), int64(2), object(12)
memory usage: 6.4+ MB

```

Duplicate Handling

A check for duplicate rows identified 534 duplicate entries. These duplicates were subsequently removed using `df.drop_duplicates(inplace=True)`, resulting in a cleaned dataset of 54,966 unique rows.

```
1 df.drop_duplicates(inplace=True)
2 print(f"Number of rows after removing duplicates: {len(df)}")

Number of rows after removing duplicates: 54966
```

Variable Identification

Columns were categorized into numerical and categorical types:

- Categorical Columns: Name, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Admission Type, Discharge Date, Medication, Test Results.
- Numerical Columns: Age, Billing Amount, Room Number.

```
1 categorical_cols = df.select_dtypes(include='object').columns
2 numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
3
4 print("Categorical Columns:", categorical_cols.tolist())
5 print("Numerical Columns:", numerical_cols.tolist())

Categorical Columns: ['Name', 'Gender', 'Blood Type', 'Medical Condition', 'Date of Admission', 'Doctor', 'Hospital', 'Insurance Provider', 'Admission Type', 'Discharge Date', 'Medication', 'Test Results']
Numerical Columns: ['Age', 'Billing Amount', 'Room Number']
```

Missing Values Check

No missing values were found across any of the columns in the dataset, eliminating the need for imputation.

```
1 print('Missing values per column:')
2 print(df.isnull().sum())

*** Missing values per column:
Name      0
Age        0
Gender     0
Blood Type 0
Medical Condition 0
Date of Admission 0
Doctor     0
Hospital   0
Insurance Provider 0
Billing Amount 0
Room Number 0
Admission Type 0
Discharge Date 0
Medication 0
Test Results 0
dtype: int64
```

Date Column Conversion

The columns Date of Admission and Discharge Date were converted from object (string) to datetime data type using `pd.to_datetime`. This is crucial for any time-based analysis, such as calculating the length of stay.

```
1 df['Date of Admission'] = pd.to_datetime(df['Date of Admission'])
2 df['Discharge Date'] = pd.to_datetime(df['Discharge Date'])
3
4 print('Data types after date conversion:')
5 display(df[['Date of Admission', 'Discharge Date']].info())
```

Data types after date conversion:

```
<class 'pandas.core.frame.DataFrame'>
Index: 54966 entries, 0 to 55499
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Date of Admission 54966 non-null  datetime64[ns]
1   Discharge Date    54966 non-null  datetime64[ns]
dtypes: datetime64[ns](2)
memory usage: 1.3 MB
None
```

Scale Analysis

- Number of Rows and Columns: 55,500 initial records and 15 columns. After removing duplicates, the dataset maintains 54,966 clean records.
- Missingness: The dataset is complete with 0% missing values, ensuring high reliability for statistical modeling.
- Sparsity or Memory Estimation: This is a dense dataset with a memory footprint of approximately 6.4 MB, allowing for high-performance processing in standard environments.

Feature Engineering

patient_feature_matrix:

Aggregated patient-level feature matrix constructed by combining demographic information with medical records. Each row represents a unique patient and summarizes their healthcare activity and characteristics.

Source Tables:

- patients
- medical_records

Join Operation:

The datasets were merged using a left join on `patient_id`:

```
# Unir `patients` con `medical_records`
patient_medical_data = pd.merge(
    patients,
    medical_records,
    on='patient_id',
    how='left'
)
```

Aggregation Level:

- Grouped by: patient_id
- Each row represents: one patient

Features Included:

patient_id	Unique patient identifier	—
age	Patient age	first
gender	Patient gender	first
blood_type	Patient blood type	first
total_admissions	Total number of admissions	count
avg_billing_amount	Average billing amount per admission	mean
avg_length_of_stay	Average hospital stay duration (days)	mean

Data Types

- Numeric: age, total_admissions, avg_billing_amount, avg_length_of_stay.
- Categorical: gender, blood_type.

Dimensions:

The feature matrix is structured so that each row represents a single patient, and each column represents a characteristic or summary of that patient's medical history.

In this case, the matrix includes:

- **Patients (rows):** each unique patient_id corresponds to one individual
- **Features (columns):** demographic attributes (such as age, gender, and blood type) and aggregated healthcare metrics (such as total admissions, average billing amount, and average length of stay)

This structure allows us to move from raw, record-level data (multiple rows per patient) to a **patient-centered view**, where all relevant information is summarized into a single row per individual.

As a result:

- The dataset becomes easier to analyze and model
- Each patient can be directly compared to others based on their characteristics
- It serves as a suitable input for machine learning tasks such as clustering, similarity analysis, and recommendation systems

Matriz de Características de Pacientes (Primeras 5 filas):

	patient_id	age	gender	blood_type	total_admissions	avg_billing_amount	avg_length_of_stay
0	0	22	Male	AB-	1	39906.147308	11.0
1	1	76	Female	A+	1	10584.185945	13.0
2	2	20	Male	O-	1	29740.960199	26.0
3	3	69	Female	B-	1	21535.554758	9.0
4	4	18	Male	B-	1	13895.551020	7.0

Dimensiones de la Matriz de Características de Pacientes: (49992, 7)

Feature Matrix Expansion

Motivation (Why the matrix was expanded)

The initial patient feature matrix was expanded to enrich each patient's representation with more descriptive and predictive attributes. The goal was to move beyond basic demographic data and incorporate information about their medical interactions, which is crucial for subsequent tasks such as clustering, segmentation, or building predictive models.

Added Features (List of added features)

Additional features added to the patient matrix included:

1. **One-Hot Encoding** for categorical variables 'Gender' and 'Blood Type'.
2. **Medical Condition Statistics**: Count of each medical condition and the most frequent condition per patient.
3. **Medication Statistics**: Count of each medication and the most frequent medication per patient.
4. **Test Results Statistics**: Proportion of 'Normal' and 'Abnormal' test results per patient.

Resulting Representation (How the final matrix turned out)

The patient feature matrix, after expansion, has 54966 rows (patients) and 33 columns (features), as shown in the code output:

Dimensiones de la Matriz de Características de Pacientes ampliada: (54966, 33)

Matriz de Características de Pacientes ampliada (Primeras 5 filas):

	patient_id	age	gender	blood_type	total_admissions	avg_billing_amount	avg_length_of_stay	Gender_Female	Gender_Male	BloodType_A+	...	most_frequent_condition
0	0	22	Male	AB-	1	39906.147308	11.0	False	True	False	...	Obesity
1	1	76	Female	A+	1	10584.185945	13.0	True	False	True	...	Arthritis
2	2	20	Male	O-	1	29740.960199	26.0	False	True	False	...	Hypertension
3	3	69	Female	B-	1	21535.554758	9.0	True	False	False	...	Diabetes
4	4	18	Male	B-	1	13895.551020	7.0	False	True	False	...	Cancer

5 rows x 33 columns

Dimensionality and Representation Analysis

Motivation

Dimensionality reduction was applied to:

- **High-dimensionality problems:** Reduce the number of features from 33 to a more manageable set.
- **Redundancy:** Eliminate redundant information among features, especially highly correlated ones.
- **Visualization:** Facilitate the representation of data in 2 or 3 dimensions to identify patterns or clusters.
- **Computational Efficiency:** Improve the performance of Machine Learning algorithms by working with a smaller number of dimensions.

Methods

PCA

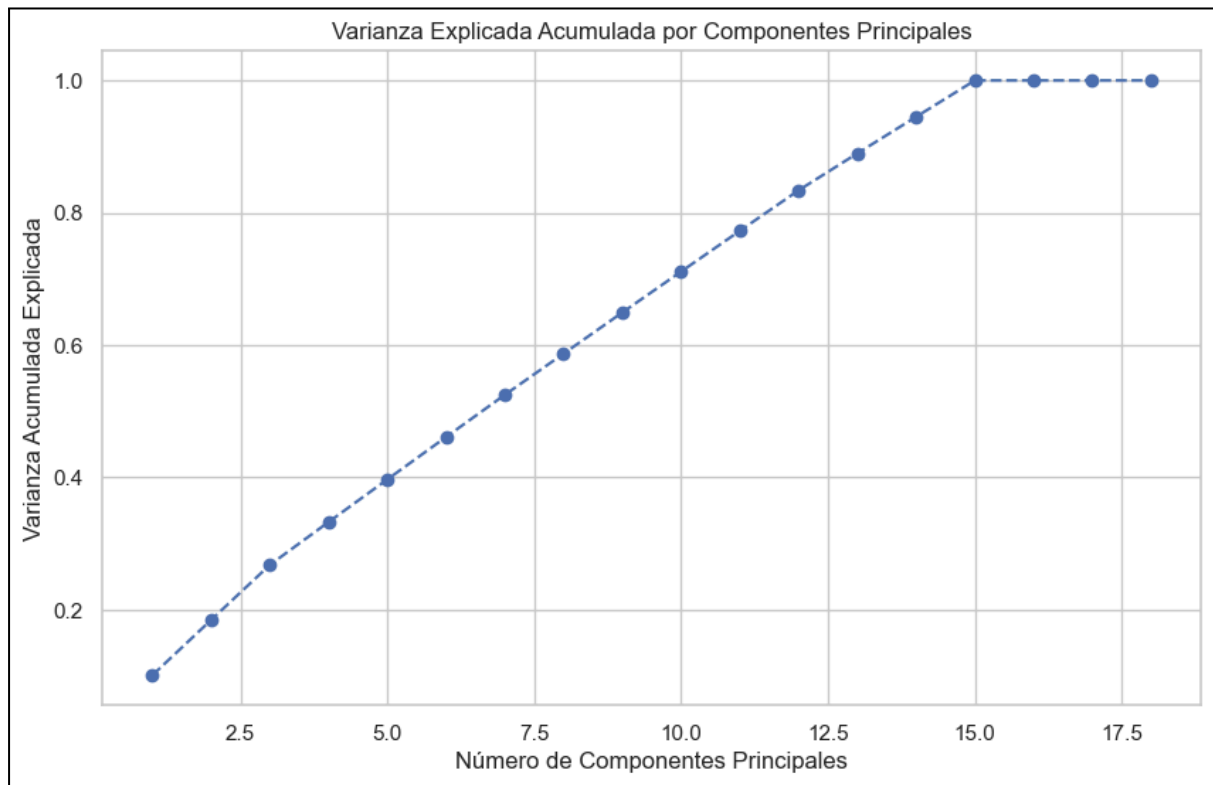
In this project, PCA was applied to the patient feature matrix after encoding categorical variables and scaling numerical features. The goal was to reduce the dimensionality of the dataset while preserving as much information as possible.

The analysis shows that variance is gradually distributed across multiple components. Approximately:

- 5 components explain around 40% of the variance
- 10 components explain around 70% of the variance
- 13 components explain around 90% of the variance

This indicates that the dataset does not have a strong low-dimensional structure, as no small number of components captures most of the information.

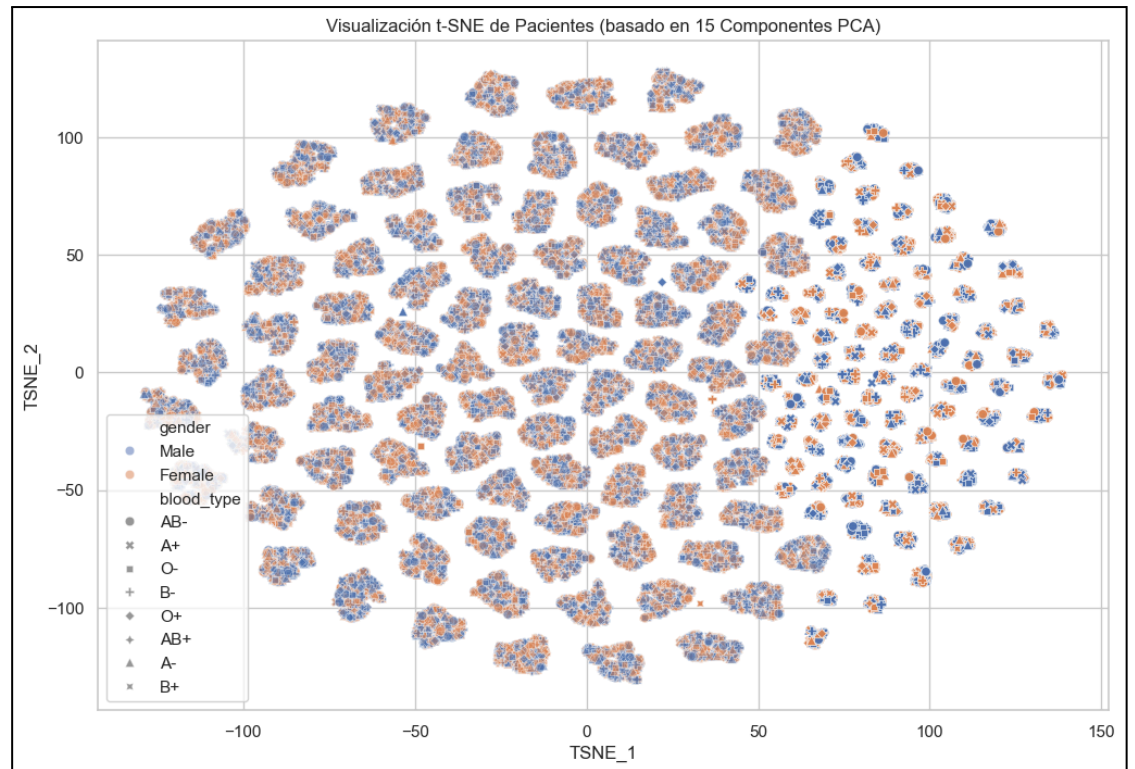
Based on this, a higher number of components (e.g., 15) was retained for downstream tasks such as t-SNE visualization, ensuring that most of the original variance is preserved.



t-SNE

In this project, t-SNE was applied to the feature matrix after dimensionality reduction using PCA. Specifically, the first principal components were used as input to t-SNE in order to reduce noise and improve the quality of the visualization.

The resulting projection provides a two-dimensional representation of patient profiles, where each point corresponds to a patient and proximity between points reflects similarity in their features.



The figure shows the t-SNE projection of patients based on the reduced feature space obtained from PCA. Each point represents a patient, and colors/shapes indicate demographic attributes such as gender and blood type.

The t-SNE projection of patients, based on the first 15 PCA components, provides a two-dimensional representation of the feature space.

- The data points form multiple **localized clusters**, indicating the presence of underlying structure in the patient feature space.
- However, these clusters are **not clearly separated**, suggesting that differences between patients are gradual rather than sharply defined.
- There is a **high overlap between genders (Male and Female)**, indicating that gender does not significantly differentiate patient profiles.
- Similarly, **blood types are widely dispersed across the plot**, with no distinct grouping patterns.

Data Preparation

For dimensionality reduction, the data was prepared as follows:

- **Encoding applied:** Categorical columns like 'Gender' and 'Blood Type' had already been transformed through One-Hot Encoding during the feature matrix expansion phase. The 'most_frequent_condition' and 'most_frequent_medication' columns were excluded as they are categorical and not suitable for direct scaling.

- **Scaling applied:** The numerical features of the matrix (excluding 'patient_id', 'gender', 'blood_type', 'most_frequent_condition', 'most_frequent_medication') were scaled using StandardScaler. This ensures that all features contribute equally to the analysis, preventing features with larger value ranges from dominating the results.
- **Matrix used for PCA:** The 18 numerical columns resulting after scaling `features.select_dtypes(include=['number'])` were used.

Results

Explained Variance

The cumulative explained variance visualization showed that:

- **15 principal components** managed to retain **100.00%** of the original variance of the scaled data.

Component Selection

A threshold was chosen to retain **95% of the variance**. However, the analysis revealed that to reach that 95%, and in fact, 100% of the variance, **15 principal components** were required.

- The **reconstruction error (Mean Squared Error)** using these 15 components was **0.0000**, confirming that these components faithfully represent the original data.

Comparison Table

Criterion	PCA (Principal Component Analysis)	t-SNE (t-Distributed Stochastic Neighbor Embedding)
Final Dimensionality	15 Components	2 Components (Visualization)
Explained Variance	100.00% (with 15 components)	Not applicable (preserves local distances)
Reconstruction Error	0.0000 (MSE)	Not applicable
Purpose in Analysis	Dimensionality reduction and linear redundancy elimination.	Visualization of groupings and non-linear relationships.
Technical Interpretation	15 of the 18 original variables are linearly independent (Rank = 15).	Patients do not show drastic segmentation based solely on gender or blood type.

Visualizations

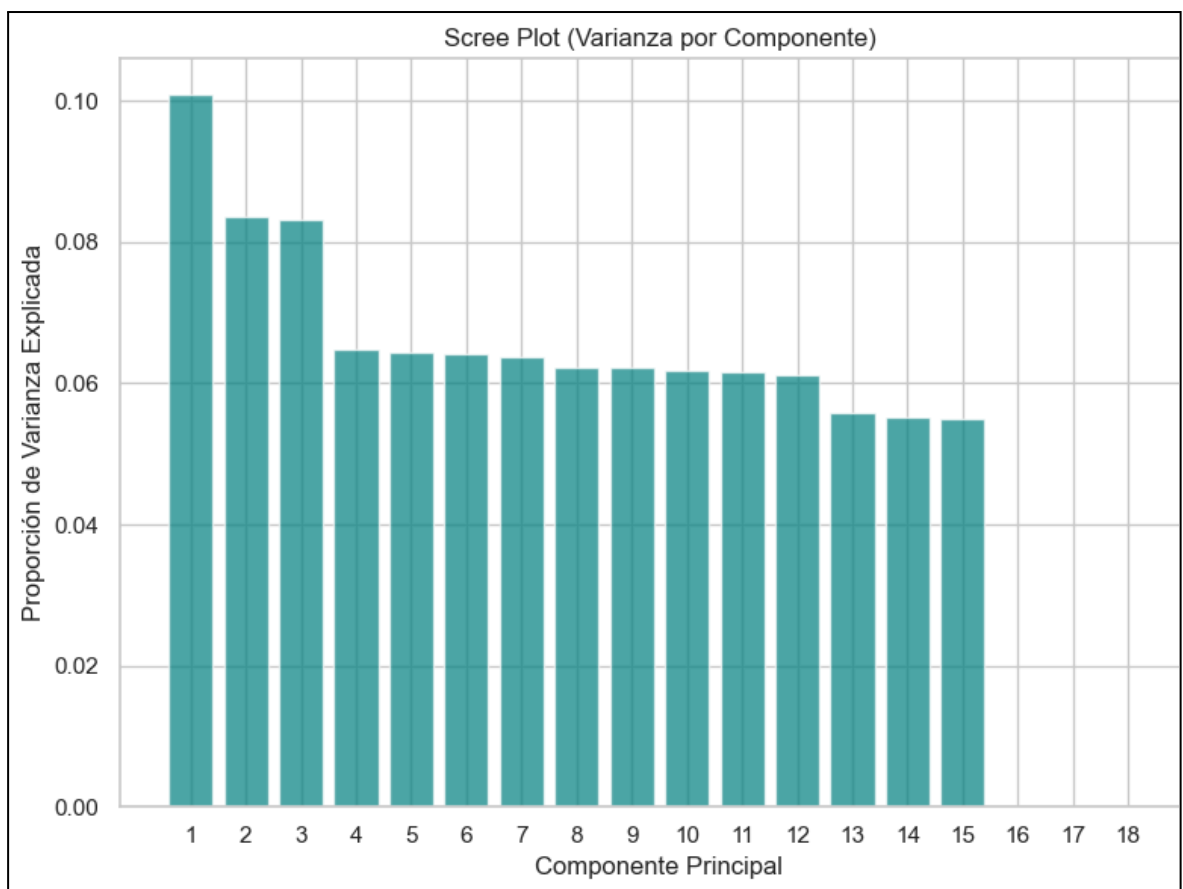
The following visualizations were generated:

- **Scree Plot :**

The scree plot displays the proportion of variance explained by each principal component in descending order.

From the plot, it can be observed that:

- The first principal component explains the largest share of variance (around 10%)
- The following components contribute slightly less, but the decrease is gradual rather than sharp
- There is no clear elbow point, indicating that variance is not concentrated in only a few components



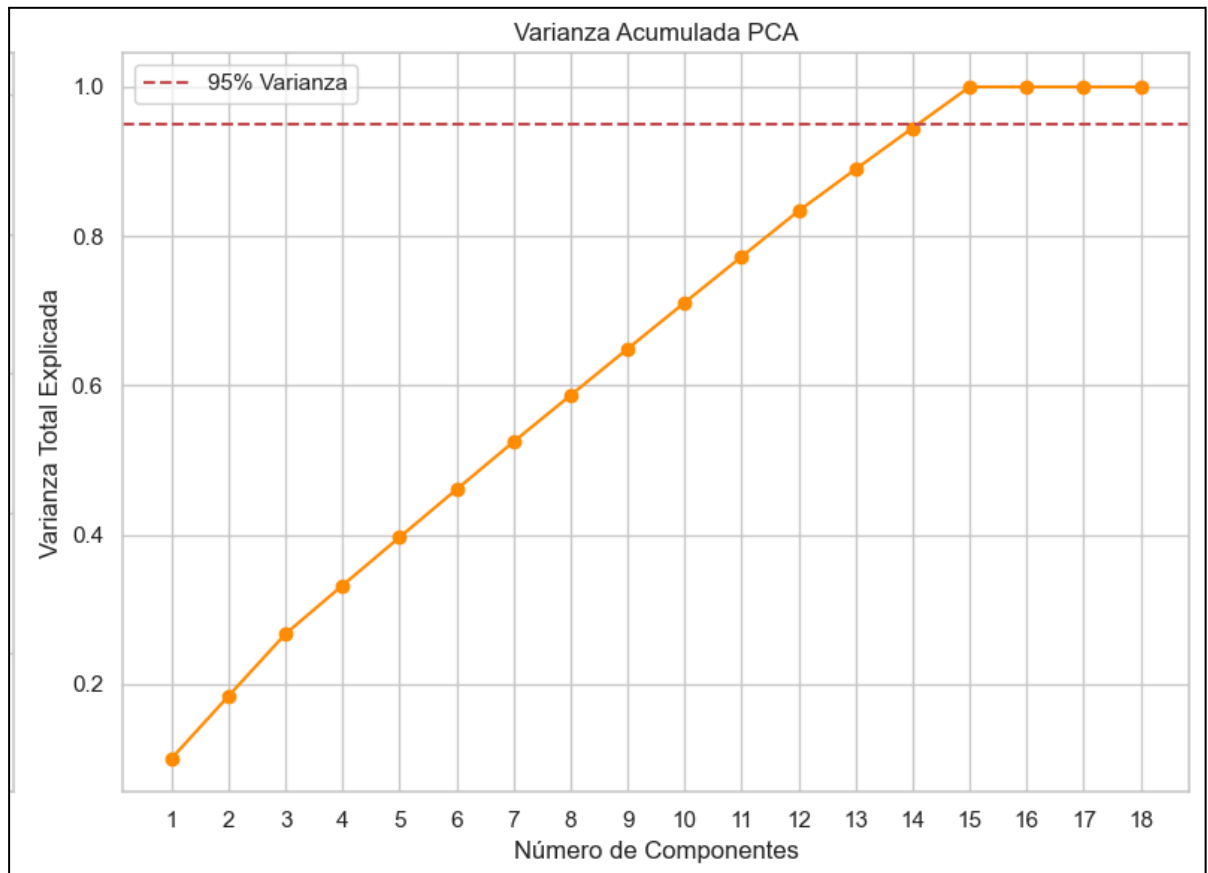
- **Cumulative PCA Variance:**

The cumulative variance plot shows how the total explained variance increases as more components are included.

Key observations:

- Around **5 components** explain **~40% of the variance**
- Around **10 components** explain **~70% of the variance**
- Around **13 components** explain **~90% of the variance**
- Around **15 components** explain **~100% of the variance**

Additionally, the 95% variance threshold is reached at approximately **14–15 components**.



Based on these results, a relatively high number of components is required to preserve the structure of the data. In this project, **15 components were retained**, as they capture nearly all the variance while still reducing dimensionality.

This selection provides a balance between:

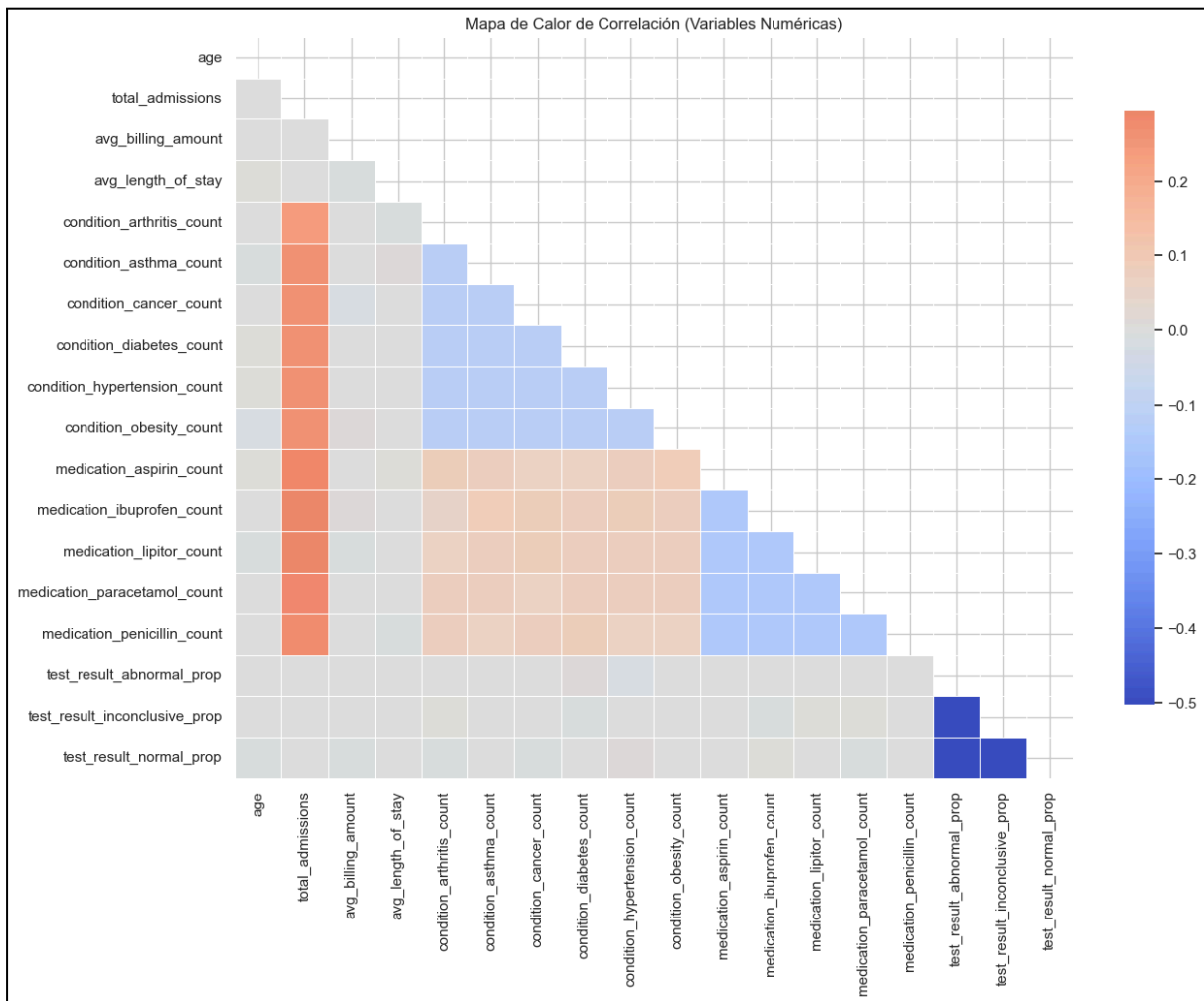
- reducing complexity
- maintaining essential information for downstream tasks such as t-SNE visualization and clustering

- **Correlation Heatmap :**

The correlation heatmap illustrates the pairwise relationships between numerical features in the patient feature matrix, using correlation coefficients ranging from -1 to 1.

- Most correlations are **weak to moderate**, indicating that features are not strongly linearly dependent.

- There is a noticeable **positive correlation among medication-related variables**, suggesting that patients who receive one type of medication are more likely to receive others.
- Similarly, **medical condition counts show mild positive correlations**, indicating some co-occurrence of conditions.
- There are **negative correlations between certain condition counts and other variables**, suggesting inverse relationships between some health conditions.
- Test result proportions (e.g., normal vs abnormal) exhibit **strong negative correlations**, which is expected since they represent complementary outcomes.



The correlation heatmap shows that while some variables are related, the overall feature space is not dominated by strong linear dependencies. This supports the use of dimensionality reduction techniques and indicates that the dataset contains diverse and informative features suitable for further analysis.

Technical Interpretation

- **Redundancy in features:** The fact that 15 principal components explain 100% of the variance of the 18 original numerical variables indicates **moderate redundancy** in the feature set. This means that not all original variables contributed completely unique information, and some of them were linearly related. PCA has consolidated this information into a more compact set.
- **Complexity of the space:** Although there was a reduction from 18 to 15 dimensions for total variance retention, the feature space remains relatively complex. However, PCA has identified the directions of greatest variance, simplifying the representation.
- **What was learned from the data:** It was learned that while there is an underlying structure that allows for information compression, demographic variables such as 'Gender' and 'Blood Type' are not the primary factors driving patient grouping or variability when visualized with t-SNE. This suggests that other characteristics (such as medical conditions, medications, or treatment patterns) have a greater weight in differentiating patient profiles.
- **Impact on clustering/recommendation:** The matrix reduced to 15 PCA components is an excellent basis for clustering algorithms or recommendation systems. It provides a denser and less noisy data representation, which can lead to:
 - **Better clusters:** More meaningful groupings by focusing on key variance.
 - **More efficient models:** Fewer features mean faster training times and less risk of overfitting.
 - **More accurate recommendations:** By relying on the most influential patient dimensions.

Repository

Repository	https://github.com/SofiaGMiranda/BIG-DATA-FINAL-PROYECT-2025-01.git
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